

Rachael Lang

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Education

University of Wisconsin - Madison

Master of Science in Computer Science

Jan 2024 – May 2025

Madison, Wisconsin

University of Wisconsin - Madison

Bachelors of Science in Computer Science, minors in Data Science, Chicano and Latino Studies

Sep 2019 – Dec 2023

Madison, Wisconsin

- Coursework: Object-Oriented Programming (Java), Data Structures and Algorithms (Java), Data Science Programming (Python), Machine Programming (C), Android App Development (Java, Kotlin), User Interfaces (JavaScript, React), Theory of Programming Languages (JavaScript, Haskell, C++, Rust), Machine Learning, Human-Computer Interaction, Computer Engineering, Discrete Mathematics, Data Modeling (R), Statistics for Engineers (R), Linear Algebra

Work Experience

Intel Corporation

Software Engineer Intern

Nov 2023 – Jan 2024

Hillsboro, Oregon

- Designed and implemented a Python script to monitor and log file modifications in a multi-user environment
- Collaborated with cross-functional teams to integrate the file change tracking script into the existing workflow
- Optimized script performance to handle large volumes of data, significantly reducing system load

Target Corporation

Software Engineer Intern

Jun 2023 – Aug 2023

Minneapolis, Minnesota

- Designed and developed responsive web application using TypeScript and React, ensuring seamless user experiences
- Engineered API endpoints using Kotlin, enabling efficient data communication between front-end and back-end systems
- Executed cross-browser compatibility testing to ensure seamless functioning of various web applications
- Conducted code reviews to ensure quality, adhering to industry best practices and promoting continuous improvement
- Contributed in an Agile ecosystem, fostering collaboration with a cross-functional team to drive software success

The J.M. Smucker Company

Software Engineer Intern

May 2022 – Aug 2022

Orrville, Ohio

- Redesigned Oracle APEX app for enhanced supply chain product tracking, improving user experience and efficiency
- Developed shell script to reduced after-hours support needs by 50% through automated task execution
- Leveraged Jaspersoft to implement an automated data export function, reducing labor-intensive manual efforts
- Conducted comprehensive testing and debugging to ensure smooth migration between legacy and replacement systems
- Collaborated closely with database administrators to optimize database performance of product tracking software

UW-Madison Division of Information Technology

Data & Metrics Team Lead

Oct 2021 – Present

Madison, Wisconsin

- Utilized Tableau to design data visualizations that optimized Help Desk operations and increased productivity
- Supervised over 100 employees, ensuring quality and efficiency for a seamless customer experience
- Applied extensive software expertise to diagnose and resolve intricate issues across 50+ products and services

Projects

Python Performance Test

- Conducted comprehensive benchmark tests independently to evaluate the performance improvements claimed by the Python Software Foundation between versions 3.10 and 3.11
- Created a concise 3-page report summarizing the processes and methodologies employed in an effective manner

Python Recipe Generator (Data Analyst)

- Transformed raw CSV data into a Python DataFrame and harnessed this dataset to develop an innovative application capable of generating over 150 distinct recipes utilizing more than 100 unique ingredients
- Guided team of 4 during a hackathon, demonstrating strong leadership by proactively assigning responsibilities and ultimately presenting our team's outstanding final product

Technical Skills

Languages: Java, Python, C/C++, HTML/CSS, JavaScript/TypeScript, R, Bash, Assembly, Haskell, Rust, Kotlin

Developer Tools: Linux, Git, Agile, Visual Studio Code, RStudio, IntelliJ, PyCharm, Android Studio